

CS5187 Final Exam Review / 期末复习资料

Course / 课程: CS5187: Vision & Image
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1. Knowledge Framework / 知识框架

Part I: Low-Level Vision / 低层视觉 (lec00-lec05)

Lecture	Topic / 主题
lec00	Introduction to Computer Vision / 计算机视觉导论
lec01	Images & Image Filtering / 图像与图像滤波
lec02	Edge Detection / 边缘检测
lec03	Image Resampling & Interpolation / 图像重采样与插值
lec04	Harris Corner Detection / Harris角点检测
lec05	Feature Invariance / 特征不变性

Part II: Geometry & Appearance / 几何与外观 (lec06-lec10, lec13-lec14)

Lecture	Topic / 主题
lec06	Feature Descriptors & Matching / 特征描述子与匹配
lec07	Image Transformations & Warping / 图像变换与变形
lec08	Image Alignment / 图像对齐
lec09	RANSAC / 随机抽样一致
lec10	Cameras & Projection / 相机与投影
lec13	Binocular Stereo / 双目立体视觉
lec14	Multi-View Stereo / 多视图立体

Part III: Recognition & Deep Learning / 识别与深度学习 (lec15-lec16)

Lecture	Topic / 主题
lec15	Neural Networks / 神经网络
lec16	CNNs, RNNs, Attention & Transformers / 卷积网络、循环网络、注意力与Transformer

Part IV: Generative Models / 生成模型 (下半学期)

Lecture	Topic / 主题
lec04	Maximum Likelihood & MLE / 最大似然估计
lec05	Latent Variable Models & ELBO / 潜变量模型与证据下界
LVM_02	VAE & Reparameterization / 变分自编码器与重参数化
NF_01	Normalizing Flow Models (1) / 标准化流模型(1)
lec08	Normalizing Flow Models (2) / 标准化流模型(2)
lec09	GANs / 生成对抗网络
lec10	f-GANs, Wasserstein GAN, BiGAN, CycleGAN / f-散度GAN, WGAN, 双向GAN, 循环GAN
lec11	Energy-Based Models / 基于能量的模型
lec12	Score Matching & NCE / 分数匹配与噪声对比估计
lec13	Score-Based Models (1) / 基于分数的模型(1)
lec14	Score-Based Models (2) / 基于分数的模型(2)
lec15	Diffusion Models (DDPM, SDE) / 扩散模型
lec16	Diffusion Models: SDE+ODE, Architecture / 扩散模型进阶

2. Core Concepts / 核心概念梳理

2.1 Image Filtering / 图像滤波

Convolution / 卷积

- Definition (定义): Replace each pixel with a weighted sum of its neighbors using a kernel. / 用卷积核将每个像素替换为其邻域的加权。
- Formula (公式): $G[i,j] = \sum H[m,n] \cdot F[i-m, j-n]$
- Key property (关键性质): Convolution is associative and commutative; convolution with self is another Gaussian. / 卷积满足结合律和交换律。

Gaussian Filter / 高斯滤波器

- Definition (定义): A low-pass filter that removes high-frequency components. / 去除高频成分的低通滤波器。
- Property (性质): Convoluting two Gaussians of σ_1 and σ_2 yields a Gaussian of $\sigma = \sqrt{(\sigma_1^2 + \sigma_2^2)}$ / 两个高斯卷积仍为高斯。

Sharpening Filter / 锐化滤波器

- Definition (定义): original + (original - blurred) = 2*original - blurred. / 原图加上(原图减模糊图)实现锐化。
- Also called (亦称): High-pass filter / 高通滤波器。

2.2 Edge Detection / 边缘检测

Image Gradient / 图像梯度

- Definition (定义): $\nabla f = [\partial f / \partial x, \partial f / \partial y]$, points in direction of most rapid increase. / 指向强度增长最快的方向。
- Gradient magnitude (梯度幅值): $\|\nabla f\| = \sqrt{(\partial f / \partial x)^2 + (\partial f / \partial y)^2}$
- Gradient direction (梯度方向): $\theta = \arctan2(\partial f / \partial y, \partial f / \partial x)$

Sobel Operator / Sobel算子

- Definition (定义): Common approximation of derivative of Gaussian. / 高斯导数的常用近似。
- Kernels (核):
 - x-direction: $\begin{bmatrix} -1, 0, 1 \\ -2, 0, 2 \\ -1, 0, 1 \end{bmatrix}$
 - y-direction: $\begin{bmatrix} 1, 2, 1 \\ 0, 0, 0 \\ -1, -2, -1 \end{bmatrix}$

Canny Edge Detector / Canny边缘检测器

- Filter image with derivative of Gaussian / 用高斯导数滤波
- Find magnitude and orientation of gradient / 求梯度幅值和方向
- Non-maximum suppression / 非极大值抑制
- Hysteresis thresholding (two thresholds: high T, low t) / 滞后阈值化
- Strong edges: $R > T$; Weak edges: $t < R < T$ (kept only if connected to strong edge) / 强边缘保留，弱边缘仅在连与强边缘时保留

Scale Space / 尺度空间

- Properties (性质): Edge position may shift with σ ; two edges may merge; an edge cannot split into two. / 边缘位置可能随 σ 偏移，可合并但不能分裂。

2.3 Image Resampling / 图像重采样

Aliasing / 混叠

- Definition (定义): Occurs when sampling rate $< 2 \times$ max frequency (Nyquist rate). / 采样率低于奈奎斯特频率时发生。
- Solution (解决方案): Gaussian pre-filtering before subsampling. / 下采样前先做高斯预滤波。

Gaussian Pyramid / 高斯金字塔

- Definition (定义): Repeatedly blur and downsample by $2\times$. / 反复模糊后2倍下采样。
- Space (空间): Total space = $4/3 \times$ original image. / 总空间为原图的4/3。

Interpolation Methods / 插值方法

- Nearest-neighbor (最近邻): Fastest, but blocky. / 最快但产生方块效应。
- Bilinear (双线性): Better quality, tent function. / 质量更好。
- Bicubic (双三次): Best quality among common choices. / 常用方法中质量最好。

2.4 Harris Corner Detection / Harris角点检测

SSD Error / SSD误差

- Formula (公式): $E(u,v) = \sum_{(x,y) \in W} w(x,y) (I(x,y,u,v) - I(x,y))^2$
- Small motion approximation (小运动近似): $E(u,v) \approx [u \ v]^T H [u \ v]$, where H is the second moment matrix. / 二阶矩矩阵近似。

Second Moment Matrix / 二阶矩矩阵

- Definition (定义): $H = 2 \sum w(x,y) [I_x^2, I_x I_y; I_x I_y, I_y^2]$
- Eigenvalues (特征值): λ_1, λ_2 determine corner/edge/flat classification. / 决定角点/边缘/平坦区域分类。

Classification / 分类

Case (情况)	λ_1, λ_2	Classification (分类)
Both small (都很小)	$\lambda_1 \approx \lambda_2 \approx 0$	Flat / 平坦区域
One large, one small (一大一小)	$\lambda_1 \gg \lambda_2$ or $\lambda_2 \gg \lambda_1$	Edge / 边缘
Both large (都很大)	$\lambda_1 \gg \lambda_2 \gg 0$	Corner / 角点

Harris Operator / Harris算子

- Formula (公式): $R = \det(H) - \alpha \cdot \text{trace}(H)^2 = \lambda_1 \lambda_2 - \alpha (\lambda_1 + \lambda_2)^2$
- Typical α (典型参数): $\alpha \in [0.04, 0.06]$

Harris Detection Steps / Harris检测步骤

- Compute image gradients / 计算图像梯度
- Square of derivatives, Gaussian filter / 梯度平方后高斯滤波
- Compute cornerness function / 计算角点响应函数
- Non-maxima suppression / 非极大值抑制

2.5 Feature Invariance / 特征不变性

Invariance vs. Equivariance / 不变性与等变性

- Invariance (不变性): Feature locations don't change after transformation. / 变换后特征位置不变。
- Equivariance (等变性): Features detected at corresponding locations in transformed images. / 变换后特征在对应位置被检测到。

Harris Invariance Properties / Harris不变性

Transformation (变换)	Property (性质)
Translation (平移)	Equivariant / 等变
Rotation (旋转)	Equivariant (eigenvalues unchanged) / 等变
Affine intensity (仿射强度)	Partially invariant / 部分不变
Scale (尺度)	NOT invariant / 不变

Scale Invariant Detection / 尺度不变检测

- Key idea (核心思想): Find local maximum in both position AND scale. / 在位置和尺度上同时寻址局部最大值。
- Laplacian of Gaussian (LoG): Blob detector, find maxima in position-scale space. / 斑点检测器，在位置-尺度空间找极大值。
- Difference of Gaussians (DoG): Approximation of LoG, used in SIFT. / LoG的近似，SIFT中使用。

Characteristic Scale / 特征尺度

- Definition (定义): Scale producing peak LoG response. / 产生LoG峰响应的尺度。

2.6 Feature Descriptors & Matching / 特征描述子与匹配

SIFT Descriptor / SIFT描述子

- Process (过程):
 - Take 16×16 window around feature / 取 16×16 窗口
 - Compute gradient orientation per pixel / 计算每个像素梯度方向
 - Threshold weak edges / 阈值化弱边缘
 - Create 4×4 grid of cells, 8 orientation bins each / 4×4 网格，每格8方向bin

- Result: 128-dimensional descriptor / 结果为128维描述子
- Properties (性质): Robust to viewpoint (up to -60°), illumination changes. / 对视点变化($<60^\circ$ 以内)和光照变化鲁棒。

HOG Descriptor / HOG描述子

- Steps: Gradient computation $\rightarrow$ Cell histograms (9 bins) $\rightarrow$ Block normalization (L2-norm). / 梯度计算 $\rightarrow$ 单元直方图 $\rightarrow$ 块归一化。

ORB Descriptor / ORB描述子

- Steps: oFAST keypoint detection $\rightarrow$ rBRIEF binary descriptor (256 bits). / oFAST关键点检测 $\rightarrow$ rBRIEF二值描述子。

Feature Matching / 特征匹配

- L2 Distance: $\|f_1 - f_2\|$
- Ratio distance: $\|f_1 - f_2\| / \|f_1 + f_2\|$ (f_2 = best match, f_2' = 2nd best) / 比值距离
- ROC Curve: True positive rate vs. false positive rate; use Area Under the Curve (AUC) indicator.

2.7 Image Transformations / 图像变换

Transformation Hierarchy / 变换层级

Transform (变换)	Matrix (矩阵)	DOF (Properties (性质))
Translation (平移)	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	2 Origin may shift / 原点可移动
Euclidean (欧氏)	$R \pm t$	3 Preserves distances / 保持距离
Similarity (相似)	$sR \pm t$	4 Preserves angles / 保持角度
Affine (仿射)	$\begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}$	6 Parallel lines preserved / 平行线保持
Projective (射影/单应)	General 3×3	8 Lines preserved, parallel not / 直线保持，平行不一定

Homography / 单应变换

- Definition (定义): 8-DOF projective transformation; defined up to scale. / 8自由度射影变换，仅尺度因式决定。
- Minimum matches (最少匹配): 4 point correspondences. / 4对匹配点。

Forward vs. Inverse Warping / 前向与逆向变形

- Forward warping: Send source pixels to target; may cause holes. / 源像素送到目标，可能产生空洞。
- Inverse warping: For each target pixel, sample from source via T^{-1} ; use interpolation. / 对每个目标像素从源采样，需插值。

2.8 Image Alignment & Least Squares / 图像对齐与最小二乘

Least Squares / 最小二乘法

- Translation (平移): $t^* = \text{mean displacement}$; normal equations: $A^T A \cdot t = A^T b$. / 平移的最优解为平均位移。
- Affine (仿射): 6 unknowns, need ≥ 3 matches, 6个未知量，至少3对匹配。
- Homography (单应): Solve via SVD: $b = \text{eigenvector of } A^T A \text{ with smallest eigenvalue}$. / 通过SVD求解。

2.9 RANSAC / 随机抽样一致

Algorithm / 算法步骤

- Randomly select s samples (minimum to fit model) / 随机选 s 个样本
- Fit model / 拟合模型
- Count inliers (within threshold ϵ) / 计数内点
- Repeat N times / 重复 N 次
- Choose model with most inliers / 选择内点最多的模型
- Refit using all inliers (least squares) / 用所有内点重新拟合

Number of Rounds / 迭代次数

- Formula (公式): $N = \lceil \log(1-p) / (\log(1-(1-\epsilon)^s)) \rceil$
- p = desired probability of success / 期望成功概率
- ϵ = outlier ratio / 外点比例
- s = minimum sample size / 最小样本数

Pros & Cons / 优缺点

- Pros: Simple, general, often works well. / 简单、通用、效果好。
- Cons: Parameters to tune; can fail for very low inlier ratios. / 需调参；内点率极低时可能失败。

2.10 Cameras & Projection / 相机与投影

Pinhole Camera / 针孔相机

- Projection (投影): $(x,y,z) \rightarrow (f \cdot x/z, f \cdot y/z)$ using similar triangles. / 利用相似三角形投影。
- Perspective projection matrix / 透视投影矩阵: $K = \begin{bmatrix} f & 0 & c_x \\ 0 & f & c_y \\ 0 & 0 & 1 \end{bmatrix}$

Camera Parameters / 相机参数

- Intrinsics (内参): focal length f , principal point (c_x, c_y) , aspect ratio α , a skew s . / 焦距、主点、长宽比、倾斜。
- Extrinsics (外参): Rotation R (3×3) and translation t (3×1). / 旋转矩阵和平移向量。
- Full projection (完整投影): $x(\text{pixel}) = K [R \mid t] X(\text{world})$

Projection Types / 投影类型

- Perspective (透视): General case, parallel lines converge. / 一般情况，平行线汇聚。
- Orthographic (正交): Distance to image plane = ∞ . $(x,y,z) \rightarrow (x,y)$. / 距离无限远。
- Weak perspective (弱透视): Scaled orthographic; good for small depth range. / 缩放正交投影。

Radial Distortion / 径向畸变

- Types: Barrel (桶形) vs. Pincushion (枕形)
- Model: $r(\text{corrected}) = r(1 + k_1 r^2 + k_2 r^4 + \dots)$

2.11 Stereo Vision / 立体视觉

Disparity & Depth / 视差与深度

- Disparity (视差): $d = x_1 - x_2$ (horizontal pixel difference between views). / 左右视图水平像素差。
- Depth (深度): $Z = f \cdot B / d$, where B = baseline, f = focal length. / 深度与视差成反比。

Stereo Matching / 立体匹配

- SSD matching: Sum of squared differences within a window. / 窗口内平方差之和。
- Window size tradeoff: Small = more detail but more noise; Large = less noise but less detail. / 窗口大小权衡。

Stereo as Energy Minimization / 立体匹配作为能量最小化

- Energy: $E(d) = Z \text{ data}_{\text{cost}}(x, y) + \lambda Z \text{ smoothness}_{\text{cost}}(d(x), d(y))$
- Smoothness costs: Potts model (0 if same, λ if different); L1 distance; / Potts模型; L1距离。

Multi-View Stereo (MVS) / 多视图立体

- SSD: Sum of SSD over multiple baselines. / 多基线SSD之和。
- Matching scores: SSD, SAD, ZNCC (zero-mean normalized cross-correlation). / 匹配评分方法。

Plane-Sweep Stereo / 平面扫描立体

- Sweep family of planes at different depths, reproject neighbors, compare. / 在不同深度扫描平面，重投影邻居并比较。
- NeRF: Represent scenes as $(x,y,z) \rightarrow \text{RGB}+\text{a}$ functions, volume rendering. / 用神经网络表示场景的辐射场。

2.12 Neural Networks / 神经网络

Linear Classifier / 线性分类器

- Score function (得分函数): $f(x, W) = Wx + b$
- Three viewpoints (三个视角): Algebraic (代数), Geometric (几何), Template matching (模板匹配)

Loss Functions / 损失函数

- Cross-entropy loss (交叉熵损失): $L = -\log(\text{exp}(f(y_i)) / \sum_j \text{exp}(f_j))$
- Softmax (软最大): $P(y=j|x) = \text{exp}(f_j) / \sum_k \text{exp}(f_k)$

Neural Network Architecture / 神经网络架构

- MLP (多层感知机): Linear functions separated by non-linear activations. / 线性函数间插入非线性激活函数。
- Gradient descent (梯度下降): $W \leftarrow W - \eta \nabla L(W)$: learning rate η crucial. / 学习率 η 至关重要。
- Backpropagation (反向传播): Efficient gradient computation via chain rule. / 通过链式法则高效计算梯度。

2.13 CNNs, RNNs, Attention, Transformers / CNN、RNN、注意力、Transformer

Convolutional Layer / 卷积层

- Output size (输出尺寸): $(N - F) / \text{stride} + 1$
- Filters (滤波器): Extend full depth of input; produce activation maps. / 滤波器覆盖输入全深度，产生激活图。
- Padding & Stride: Control output spatial dimensions. / 控制输出空间维度。

Pooling Layer / 池化层

- Max pooling (最大池化): Reduces spatial dimensions; operates per activation map. / 减小空间维度，逐激活图操作。

RNN / 循环神经网络

- Recurrence (递推): $h_t = f(W \cdot x_t + U \cdot h_{t-1})$: same weights at each time step. / 每个时间步使用相同权重。
- BPTT: Backpropagation Through Time. / 时间反向传播。
- Truncated BPTT: Backpropagate only for limited steps. / 截断时间反向传播。
- Types: Vanilla RNN, LSTM, GRU. / 类型。

Self-Attention / 自注意力

- Query-Key-Value (查询-键-值): Attention(Q,K,V) = softmax(QK^T / $\sqrt{d_k}) \cdot V$
- Multi-Head Attention (多头注意力): Multiple attention heads in parallel; concatenate outputs. / 多个注意力头并行，拼接输出。

Transformer

- Encoder (编码器): Self-attention + FFN, with residual connections and layer norm. / 自注意力+前馈网络，带残差连接和层归一化。
- Decoder (解码器): Masked multi-head attention (cannot see future) + cross-attention + FFN. / 掩码多头注意力+交叉注意力+前馈网络。
- Positional Encoding (位置编码): Injects sequence order information. / 注入序列顺序信息。

2.14 Maximum Likelihood Estimation / 最大似然估计

KL Divergence / KL散度

- Definition (定义): $D_{\text{KL}}(p \| q) = E_{x \sim p}[\log p(x)/q(x)] \geq 0$; asymmetric. / 非对称，衡量两个分布的差异。
- Interpretation (解释): Extra bits needed when using code optimized for q on data from p . / 用 q 的编码压缩来自 p 的数据时多用的比特数。

MLE / 最大似然估计

- Formula (公式): $\theta^{\text{ML}} = \arg\max_{\theta} -\mathcal{D}_{\text{KL}}(\theta \| D)$ / $\log p(\theta)$

- Equivalence (等价性): Minimizing $\text{KL}(p_{\text{data}} \| p_{\theta}) \equiv$ maximizing expected log-likelihood. / 最小化KL散度等价于最大化期望对数似然。

Monte Carlo Estimation / 蒙特卡罗估计

- Unbiased (无偏): $E[g(\theta)] = E[g(x)]$
- Variance (方差): $\text{Var}(g) = \text{Var}(g(x)/N)$; reduces with more samples. / 更多样本降低方差。

Bias-Variance Tradeoff / 偏差-方差权衡

- High bias (高偏差): Underfitting (model too simple). / 欠拟合。
- High variance (高方差): Overfitting (model too complex). / 过拟合。
- Regularization (正则化): objective = loss + $R(M)$. / 目标函数加正则项。

2.15 Latent Variable Models & VAE / 潜变量模型与变分自编码器

Latent Variable Models / 潜变量模型

- Generative process (生成过程): $z \sim p(z)$, $x = p(x|z)$. / 先采样潜变量，再由其生成数据。
- Marginal likelihood (边际似然): $p(x;\theta) = \int p(x,z;\theta) dz$ — often intractable. / 通常难以计算。

Mixture of Gaussians / 高斯混合模型

- $z \sim \text{Categorical}(1, \dots, K)$; $p(x|z=k) = N(\mu_k, \Sigma_k)$. / 离散潜变量的混合模型。

VAE / 变分自编码器

- Infinite mixture (无限混合): $z \sim N(0,1)$; $p(x|z) = N(\mu_{\theta}(z), \Sigma_{\theta}(z))$; $\mu_{\theta}, \Sigma_{\theta}$ are neural networks. / 连续潜变量的无限混合高斯。

ELBO / 证据下界

- Formula (公式): $\log p(x;\theta) \geq \mathbb{E}_q(q(z;\phi)) \log p(x,z;\theta) + H(q) = L(z;\phi;\theta)$
- Equality condition (等号条件): When $q(z;\phi) = p(z|x;\theta)$, i.e., $\text{KL}(q \| p|x;\theta)=0$. / 当变分分布等于真实后验时等号成立。
- Decomposition (分解): $\log p(x) = \text{ELBO} + D_{\text{KL}}(q(z;\phi) \| p(z|x;\theta))$

Autoencoder Perspective / 自编码器视角

- ELBO = Reconstruction term - KL term: $L = E_{q(z;\phi)}[\log p(x|z;\theta)] - D_{\text{KL}}(q_{\phi}(z) \| p(z))$
- Reconstruction (重建项): Encourages $x \approx \hat{x}$. / 鼓励重建接近原始输入。
- KL term (KL项): Forces latent distribution close to prior $p(z)$. / 迫使潜变量分布接近先验。

Reparameterization Trick / 重参数化技巧

- Key idea (核心思想): Sample $e \sim N(0,1)$, then $z = \mu + \sigma e$. / 将随机性转移到 e ，使梯度可通过确定性函数传递。
- Benefit (优势): Enables gradient-based optimization; lower variance than REINFORCE. / 实现梯度优化，方差低于REINFORCE。

Amortized Inference / 摊销推断

- Idea (思想): Learn $q_{\phi}(z|x) = \text{encoder}(x)$ to map each x to variational parameters. / 学习编码器网络将 x 映射到变分参数，避免逐点优化。

2.16 Normalizing Flow Models / 标准化流模型

Core Idea / 核心思想

- Lipschitz constraint enforcement (Lipschitz约束实现):
- Weight clipping (权重裁剪): Clamp weights to $[-c, c]$. Simple but can cause optimization issues. / 简单但可能导致优化问题。
- Gradient penalty (梯度惩罚, WGAN-GP): Add $\lambda \cdot \mathbb{E}(\| \nabla_{\mathbf{x}} D(\mathbf{x}) \|_2 - 1)^2$ to the objective. 梯度惩罚, / 稳定的梯度惩罚方法。
- Advantages over original GAN (相比原始GAN的优势):
- Loss correlates with sample quality / 损失与样本质量相关
- More stable training / 训练更稳定
- No mode collapse in practice / 实践中无模式坍塌

2.20 BiGAN / 双向生成对抗网络

Bidirectional GAN (BiGAN) / 双向GAN

- Motivation (动机): Standard GANs learn a generator $G: \mathbf{z} \rightarrow \mathbf{x}$ but cannot infer the latent representation $\mathbf{z}$ given $\mathbf{x}$. / 标准GAN只学到生成器, 无法推断潜在变量表示。
- Architecture (架构): Add an encoder $E: \mathbf{x} \rightarrow \mathbf{z}$ alongside the generator $G: \mathbf{z} \rightarrow \mathbf{x}$; discriminator D takes $(\mathbf{x}, \mathbf{z})$ pairs and distinguishes $\mathbf{z}$. $E(\mathbf{x})$ from $(G(\mathbf{z}), \mathbf{z})$. / 增加编码器 E , 判断器在 $(\mathbf{x}, \mathbf{z})$ 对上区分真假样本与生成对。
- Objective (目标): $\min_{\mathbf{G}, \mathbf{E}} \max_{\mathbf{D}} \mathbb{E} [D(\mathbf{Y}, \mathbf{E}, \mathbf{G}) = E_{\mathbf{x}}(\mathbf{x} \sim p_{\text{data}}) \log D(\mathbf{x}, E(\mathbf{x})) + E_{\mathbf{z}}(\mathbf{z} \sim p_{\mathbf{z}}) \log(1 - D(G(\mathbf{z}), \mathbf{E}(\mathbf{G}(\mathbf{z}))))]$
- Result (结果): At equilibrium, $E = G^{-1}$ (encoder inverts the generator). / 均衡时编码器为生成器的逆。
- ALI (Adversarially Learned Interference): Simultaneous proposal, equivalent to BiGAN. / 同时提出的约束方法。

2.21 CycleGAN / 循环生成对抗网络

Image-to-Image Translation / 图像到图像翻译

- Problem (问题): Given images from two domains X and Y , learn to translate between them. / 给定两个域的图像, 学习在域间翻译。
- Paired vs. Unpaired (配对与非配对): Paired examples are expensive; CycleGAN works with unpaired data. / 配对数据昂贵, CycleGAN使用非配对数据。
- CycleGAN Architecture / CycleGAN架构:
 - Generators (生成器): $G_X: Y \rightarrow X$ and $G_Y: X \rightarrow Y$, 双向生成器。
 - Discriminators (判别器): $D_X(Y)$ distinguishes real X from $G_Y(Y)$. / 判别器。
 - Adversarial loss (对抗损失): $\mathcal{L}_{\text{GAN}}(G, D, Y, X, Y) + \mathcal{L}_{\text{GAN}}(F, D, X, X, Y)$. / 双向对抗损失。
 - Cycle consistency loss (循环一致性损失): $\lambda(E_X(X) \| F(G(X)) - X \|) + \mathcal{E}_Y(Y) \| G(F(Y)) - Y \|$. / 确保往返翻译一致。
 - Overall objective (总目标): $\min_{\mathbf{G}, \mathbf{F}} \max_{\mathbf{D}_X, \mathbf{D}_Y} \mathcal{L}_{\text{GAN}} + \lambda \cdot \mathcal{L}_{\text{cyc}}$.

AlignFlow / 流对齐

- Key idea (核心思想): If G is a normalizing flow model, $F = G^{-1}$ is free; exactly cycle-consistent. / 若生成器为流模型, 逆方向自然可得, 精确循环一致。
- Training (训练): Can train via MLE and/or adversarial learning. / 可通过MLE和/或对抗学习训练。

StarGAN / 星形GAN

- Extension (扩展): Handles multiple domains with a single generator. / 单一生成器处理多个域的损失。

2.22 Energy-Based Models / 基于能量的模型

EBM Formulation / EBM公式

- Definition (定义): $p_{\theta}(\mathbf{x}) = \exp(f_{\theta}(\mathbf{x})) / Z(\theta) = (1 / Z(\theta)) \exp(f_{\theta}(\mathbf{x}))$, where $f_{\theta}(\mathbf{x})$ is any function (e.g., neural network) and $Z(\theta) = \int \exp(f_{\theta}(\mathbf{x})) d\mathbf{x}$ is the partition function. / 任意函数经指数后归一化, 分母为配分函数。
- Key property (关键性质): Very flexible architecture – f_{θ} can be any neural network (no invertibility, factorization, or latent variable constraints). / 架构极其灵活, f_{θ} 可为任意神经网络。
- Partition function (配分函数): $Z(\theta) = \int \exp(f_{\theta}(\mathbf{x})) d\mathbf{x}$ – generally intractable. / 通常不可计算。

Energy Function / 能量函数

- Interpretation (解释): $-f_{\theta}(\mathbf{x})$ is the “energy”; low energy = high probability. / 负 f_{θ} 为“能量”, 低能量对应高概率。
- Non-negativity trick (非负技巧): Given any function f_{θ} , ensure non-negativity via $p_{\theta}(\mathbf{x}) = \exp(f_{\theta}(\mathbf{x})) / Z(\theta)$, if $f_{\theta}(\mathbf{x}) \leq 0$, or $\log(1 + \exp(f_{\theta}(\mathbf{x})))$. / 通过平方、指数、绝对值等保证非负。

Maximum Likelihood Training / 最大似然训练

- Log-likelihood (对数似然): $\log p_{\theta}(\mathbf{x}) = f_{\theta}(\mathbf{x}) - \log Z(\theta)$. / 对数似然等于函数值减去配分函数对数。
- Gradient (梯度): $\nabla_{\theta} \log p_{\theta}(\mathbf{x}_{\text{data}}) = \nabla_{\theta} f_{\theta}(\mathbf{x}_{\text{data}}) - \mathbb{E}(\mathbf{x} \sim p_{\theta}) [\nabla_{\theta} f_{\theta}(\mathbf{x})]$. / 梯度=正相值 (推高训练数据能量) 减负相位 (拉低模型采样能量)。
- Positive phase vs. Negative phase (正相位与负相位):
- Positive phase (正相位): $\nabla_{\theta} f_{\theta}(\mathbf{x}_{\text{data}})$ – increases probability of training data. / 提高训练数据概率。
- Negative phase (负相位): $\mathbb{E}(\mathbf{x} \sim p_{\theta}) [\nabla_{\theta} f_{\theta}(\mathbf{x})]$ – decreases probability of model samples. / 降低模型样本概率。

Contrastive Divergence / 对比散度

- Algorithm (算法): Instead of computing $Z(\theta)$ exactly, use Monte Carlo estimate. / 用蒙特卡洛估计代替精确计算配分函数。
- Update step (更新步骤): $\nabla_{\theta} f_{\theta}(\mathbf{x}_{\text{data}}) - \nabla_{\theta} f_{\theta}(\mathbf{x}(\text{sample}))$, where $\mathbf{x}(\text{sample}) \sim p_{\theta}$. / 训练数据的梯度近似负相位。
- Key idea (核心思想): Make training data more likely than typical samples from the model. / 使训练数据比模型典型数据概率更高。

Sampling from EBMs / 从EBM采样

- MCMC (马尔可夫链蒙特卡洛):
- Initialize $\mathbf{x}$ randomly / 随机初始化
- Propose $\mathbf{x}' = \mathbf{x} + \epsilon$ and / 提议新样本
- If $f_{\theta}(\mathbf{x}') > f_{\theta}(\mathbf{x})$: accept: else accept with probability $\exp(f_{\theta}(\mathbf{x}') - f_{\theta}(\mathbf{x}))$ / 接受或以概率接受
- Converges to p_{θ} . but can be slow / 收敛但可能很慢
- Langevin MCMC (朗之万MCMC): $\mathbf{x}_{t+1} = \mathbf{x}_t + \epsilon \nabla_{\theta} f_{\theta}(\mathbf{x}_t) + \sqrt{2\epsilon} \mathbf{z}_t$, where $\mathbf{z}_t \sim \mathcal{N}(0, \mathbf{I})$. / 利用梯度信息独立采样的MCMC方法。
- Score function (分数函数): For EBMs, $\nabla_{\theta} \log p_{\theta}(\mathbf{x}) = \nabla_{\theta} f_{\theta}(\mathbf{x})$ (since $\nabla_{\theta} \log Z(\theta) = 0$). / EBM的分数函数 (对数密度对 θ 的梯度) 恰好等于 f_{θ} 的梯度。

Score-Based Generative Modeling / 基于分数的生成模型

- Key Idea (核心思想): Directly estimate the score function $\nabla_{\theta} \log p(\mathbf{x})$ (using a neural network $s_{\theta}(\mathbf{x})$) $\approx \nabla_{\theta} \log p(\mathbf{x})$. / 直接用神经网络估计分数函数。
- Score matching (分数匹配): Train s_{θ} by minimizing $\mathbb{E}(\| p_{\theta}(\text{data}) \|_2 - s_{\theta}(\mathbf{x}) - \nabla_{\theta} \log p_{\theta}(\text{data}) \|^2)$ (or variants avoiding $\nabla_{\theta} \log p_{\theta}(\text{data})$). / 通过分数匹配训练。
- Denoising Score Matching (去噪分数匹配): Add noise σ to data, then match the score of the perturbed distribution. / 对数据加噪后匹配扰动分布的分数。
- Noise Conditional Score Network (NCSN): Use multiple noise scales $\sigma_1 > \sigma_2 > \dots > \sigma_L(\mathbf{L})$: train one network conditioned on noise level. / 使用多尺度噪声, 训练条件分数网络。
- Annealed Langevin Dynamics (退火朗之万动力学): Sequentially sample with decreasing noise levels; each level initialized from previous. / 逐步降低噪声水平采样。

EBM vs. Other Generative Models / EBM与其他生成模型比较

- vs. Autoregressive/Flow/VAE: EBM has much more flexible architecture (no restrictions on f_{θ}), but sampling is slower (requires MCMC). / EBM架构更灵活, 但采样需要MCMC。
- vs. GAN: EBM provides stable training and tractable (unnormalized) likelihood; GAN has fast sampling but unstable training and intractable likelihood. / EBM训练稳定, GAN训练快速但训练不稳定。
- Composability (可组合性): $p_{\theta}(\text{combined}) \propto \Pi p_{\theta_i}(\mathbf{x})^{\alpha_i}$ – can combine multiple EBMs by multiplying their densities. / 多个EBM可线性组合, 适用于零样本组合推理。

3. Key Formulas & Theorems / 重点公式与定理

3.1 Convolution / 卷积

Formula: $F[k, j] = \sum_z \sum_n H[m, n] \cdot F[i, m, j-n]$
Variables: H = kernel, F = input image, G = output image
Conditions: Kernel size determines neighborhood / 核大小决定邻域范围

3.2 Gaussian Filter / 高斯滤波

Formula: $G(\mathbf{x}, \sigma, \mathbf{y}) = (1 / (2\pi\sigma^2)) \exp(-(x-y)^2 / (2\sigma^2))$
Variables: σ = standard deviation (scale) / 标准差 (尺度)
Conditions: Larger σ = more blurring / σ 越大越模糊

3.3 Image Gradient / 图像梯度

Formula: $\nabla f = [\partial f / \partial x, \partial f / \partial y]^T$
Variables: $\partial f / \partial x, \partial f / \partial y$ = partial derivatives / 偏导数
Conditions: Computed via finite differences or derivative of Gaussian / 通过有限差分或高斯导数计算

3.4 Harris Corner Response / Harris角点响应

Formula: $R = \det(H) = \lambda_1 \cdot \text{trace}(H)^2 - \lambda_2 \cdot \lambda_3 - \alpha(\lambda_1 + \lambda_2)^2$
Variables: H = second moment matrix, $\alpha \in [0.04, 0.06]$, λ_1, λ_2 = eigenvalues
Conditions: $R > 0$ → corner; $R \approx 0$ → edge; $R < 0$ → flat (using alternate sign convention)

3.5 Perspective Projection / 透视投影

Formula: $(x_p, y_p) = (fx(z) + c_x, fy(z) + c_y) / z$
Variables: f = focal length, $p(x, y)$ = principal point
Conditions: Pinhole camera model / 针孔相机模型

3.6 Stereo Depth / 立体深度

Formula: $Z = f \cdot B / d$
Variables: Z = depth, f = focal length, B = baseline, d = disparity
Conditions: Rectified stereo pair / 校正后的立体对

3.7 RANSAC Iterations / RANSAC迭代次数

Formula: $N = \lceil \log(1-p) / \log(1-(1-e)^{s(n)}) \rceil$
Variables: p = success probability, e = outlier ratio, s = min sample size
Conditions: Assumes $< 50\%$ outliers / 假设外点比例 $< 50\%$

3.8 ELBO / 证据下界

Formula: $\log p(\mathbf{x} | \theta) \geq \mathbb{E}_q(\log p(\mathbf{x} | \theta)) - \mathbb{E}_q(\log q(\mathbf{z} | \phi)) = \mathbb{E}_q(\log p(\mathbf{x} | \theta, \mathbf{z})) - D_{\text{KL}}(q(\mathbf{z} | \phi) \parallel p(\mathbf{z}))$
Variables: q = variational distribution, θ = model params, ϕ = variational params
Conditions: Equality when $q(\mathbf{z} | \phi) = p(\mathbf{z} | \theta)$ / 当变分分布等于后验时等号成立

3.9 Change of Variables (Jacobian / 变量替换 (标准化))

Formula: $p_Y(\mathbf{y}) = p_X(Z(f^{-1}(\mathbf{y}))) \cdot |\det(D f^{-1}(\mathbf{y}))|$
Variables: f = invertible transformation, $\mathbf{x}, \mathbf{z} \in \mathbb{R}^n$ (same dimension)
Conditions: f must be invertible; $\mathbf{z}$: continuous / f 必须可逆; $\mathbf{z}$:连续且同维

3.10 GAN Objective / GAN目标

Formula: $\min_{\mathbf{G}} \max_{\mathbf{D}} \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}} [\log(1 - D(G(\mathbf{z})))]$
Variables: G = generator, D = discriminator, $p_{\mathbf{z}}$ = prior on $\mathbf{z}$

Conditions: At optimum, equivalent to minimizing JSD / 最优时等价于最小化JS散度

3.11 Cross-Entropy Loss / 交叉熵损失

Formula: $L = -\log(\exp(f_{\mathbf{y}}(\mathbf{x})) / \sum_i \exp(f_i))$
Variables: f_i = score for class i , y_i = true label

Conditions: Most common classification loss / 最常用的分类损失

3.12 Conv Output Size / 卷积输出尺寸

Formula: $\text{output} = (N - F + 2P) / \text{stride} + 1$
Variables: N = input size, F = filter size, P = padding
Conditions: Must be integer / 必须为整数

3.13 F-Divergence / F-散度

Formula: $D_f(p, q) = \mathbb{E}_{\mathbf{x} \sim q} [f(p(\mathbf{x}) / q(\mathbf{x}))]$, where f is convex with $f(1) = 0$
Variables: p, q = probability distributions, f = convex generator function
Conditions: $D_f(p, q) \geq 0$; equals 0 iff $p = q$ (非负, 等号当且仅当两分布相同)

3.14 Wasserstein Distance / Wasserstein距离

Formula: $W(p, q) = \inf_{\gamma} \mathbb{E}[\sum_{i=1}^n |x_i - y_i|]$ (1-D) ≤ 1 (1-D) ≤ 1 (1-D) ≤ 1 (1-D)
Variables: p, q = real and generated distributions, f = Lipschitz function
Conditions: Meaningful even for non-overlapping distributions / 即使分布不重叠也有意义

3.15 EBM Probability / EBM概率

Formula: $p_{\theta}(\mathbf{x}) = (1 / Z(\theta)) \exp(f_{\theta}(\mathbf{x}))$, where $Z(\theta) = \int \exp(f_{\theta}(\mathbf{x})) d\mathbf{x}$
Variables: $f_{\theta}(\mathbf{x})$ = any function (energy/negative), $Z(\theta)$ = partition function
Conditions: $Z(\theta)$ generally intractable / 配分函数通常不可计算

3.16 EBM Log-Likelihood Gradient / EBM对数似然梯度

Formula: $\nabla_{\theta} \log p_{\theta}(\mathbf{x}_{\text{data}}) = \nabla_{\theta} f_{\theta}(\mathbf{x}_{\text{data}}) - \mathbb{E}(\mathbf{x} \sim p_{\theta}) [\nabla_{\theta} f_{\theta}(\mathbf{x})]$
Variables: First term = positive phase, second term = negative phase
Conditions: Negative phase estimated via MCMC samples / 负相位通过MCMC样本估计

3.17 Langevin MCMC / 朗之万MCMC

Formula: $\mathbf{x}_{t+1} = \mathbf{x}_t + \epsilon \nabla_{\theta} f_{\theta}(\mathbf{x}_t) + \sqrt{2\epsilon} \mathbf{z}_t$, where $\mathbf{z}_t \sim \mathcal{N}(0, \mathbf{I})$
Variables: ϵ = step size, $\nabla_{\theta} f_{\theta}$ = score function
Conditions: Converges to p_{θ} as $\epsilon \rightarrow 0$, $T \rightarrow \infty$ (步长趋于0, 迭代次数趋于无穷时收敛)

4. Typical Questions & Solutions / 典型问题与解答

Q1: Gaussian Filter Properties / 高斯滤波器性质

Problem (问题): If you convolve an image with a Gaussian of σ_1 and then with a Gaussian of σ_2 , what is the effective σ ?
Solution (解答): $\sigma_{\text{eff}} = \sqrt{\sigma_1^2 + \sigma_2^2}$. This follows from the property that convolution of two Gaussians is another Gaussian, and their variances add. / 两个高斯卷积仍为高斯, 方差相加。

Q2: Nyquist Rate / 奈奎斯特率

Problem (问题): An image has maximum frequency 50 cycles/pixel. What is the minimum sampling rate to avoid aliasing?
Solution (解答): Nyquist rate = $2 \times \text{max frequency} = 100$ samples/pixel interval. Sampling rate must be ≥ 100 . / 采样率必须至少为最高频率的两倍。

Q3: Harris Corner Classification / Harris角点分类

Problem (问题): For a window where the second moment matrix H has eigenvalues $\lambda_1 = 5000$ and $\lambda_2 = 50$, what type of region is this?
Solution (解答): $\lambda_1 \gg \lambda_2 \rightarrow$ This is an edge region. The intensity changes significantly in one direction but not the other. / 这是一个边缘区域, 强度在一个方向变化大但在另一方向变化小。

Q4: Homography DOF / 单应变换自由度

Problem (问题): How many point correspondences are needed to solve for a homography?
Solution (解答): A homography has 8 DOF (3x3 matrix up to scale). Each point correspondence gives 2 equations. Therefore, minimum 4 point correspondences are needed. / 单应矩阵有8个自由度, 每对匹配提供2个方程, 至少需4对。

Q5: RANSAC Rounds / RANSAC轮数计算

Problem (问题): For computing a homography (s=4) with 30% outliers and p=0.99, how many RANSAC rounds?
Solution (解答): $N = \lceil \log(1-0.99) / \log(1-(1-0.3)^4) \rceil = \log(0.01) / \log(1-0.7^4) \approx 17$ rounds. / 约需17轮迭代。

Q6: Stereo Depth Calculation / 立体深度计算

Problem (问题): Given a stereo system with baseline $B=0.2\text{m}$, focal length $f=500$ pixels, and disparity $d=50$ pixels, what is the depth Z ?
Solution (解答): $Z = f \cdot B / d = 500 \times 0.2 / 50 = 2.0$ meters. / 深度为2米。

Q7: Conv Layer Output Size / 卷积层输出尺寸

Problem (问题): Input 32x32x3, 10 filters of size 5x5x3, stride 1, padding 0. What is the output size?
Solution (解答): $\text{Output} = (32-5+0)/1 + 1 = 28$. So output size is $28 \times 28 \times 10$. / 输出尺寸为 $28 \times 28 \times 10$ 。

Q8: VAE ELBO Decomposition / VAE的ELBO分解

Problem (问题): Write the ELBO in terms of reconstruction loss and KL divergence.
Solution (解答): $\text{ELBO} = \mathbb{E}_{q(\mathbf{z}|\mathbf{x})} [\log p(\mathbf{x}|\mathbf{z})] - D_{\text{KL}}(q(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}))$. First term: reconstruction loss (autoencoding); Second term: KL divergence of approximate posterior from prior / 近似后验与先验的KL散度。

Q9: Flow Model Likelihood / 标准流似然计算

Problem (问题): In a normalizing flow with 3 invertible layers, write the log-likelihood.
Solution (解答): $\log p(\mathbf{X}) = \log p(\mathbf{Z}) (f'(\mathbf{x})) + \sum_{m=1}^3 \log |\det(D f_m)|$ (or $\sum \log |J|$) where $\mathbf{z}_m = f'_m(\mathbf{x})$ and $\mathbf{z}_m = \mathbf{z}_{m-1} f'_m(\mathbf{z}_{m-1})$. / 对数似然等于先验对数概率加上各层雅可比行列式对数之和。

Q10: GAN Optimal Discriminator / GAN最优判别器

Problem (问题): Derive the optimal discriminator for fixed generator.
Solution (解答): $D^*(\mathbf{x}) = (p_{\text{data}}(\mathbf{x}) / (p_{\text{data}}(\mathbf{x}) + p_{\text{G}}(\mathbf{x})))$. When $p_{\text{G}}(\mathbf{x}) = p_{\text{data}}(\mathbf{x})$: $D^* = 1/2$ (cannot distinguish) / 无法区分真假 - When $p_{\text{G}}(\mathbf{x}) \neq p_{\text{data}}(\mathbf{x})$: D^* assigns higher probability to whichever distribution has higher density at $\mathbf{x}$. / 判别器给密度更高的分布更高的概率。

Q11: Reparameterization Trick / 重参数化技巧

Problem (问题): Why can't we directly backpropagate through sampling $\mathbf{z} \sim q(\mathbf{z})$?
Solution (解答): Sampling is a stochastic operation – gradients cannot flow through random nodes. Reparameterization: $\mathbf{z} = \mu + \sigma \epsilon$ where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$. Now $\mathbf{z}$ is a deterministic function of μ and σ , and gradients can flow through μ and σ . / 采样是随机操作, 梯度无法通过。重参数化将随机性转移到 ϵ , 使梯度可通过确定函数 μ 和 σ 传递。

Q12: Change of Variables (1D) / 变量替换(一维)

Problem (问题): If $Z \sim \mathcal{U}[0, 2]$ and $X = 4Z$, find $p_X(x)$.
Solution (解答): $p_X(x) = 1/4$, $h'(0) = 1/4$, $p_X(x) = 1/4$ (1-D) $\rightarrow 1/2$ (1-D) $\rightarrow 1/4$ = $1/8$ for $x \in [0, 8]$. This is uniform on $[0, 8]$, as expected. / 结果在 $[0, 8]$ 上的均匀分布。

Q13: NICE vs Real-NVP / NICE与Real-NVP区别

Problem (问题): What is the key difference between NICE and Real-NVP coupling layers?
Solution (解答): - NICE: $\mathbf{x}_{t+1}(\mathbf{x}) = \mathbf{z}(d+1-n) + \mathbf{m}_{\text{t}} \mathbf{b}_{\text{t}}(1-d)$; $\det(J) = 1$ (volume-preserving). / 加性耦合, 体积保持。 - Real-NVP: $\mathbf{x}_{t+1}(d+1-n) = \mathbf{z}(d+1-n) \odot \exp(\mathbf{a}_{\text{t}} \mathbf{b}_{\text{t}}(1-d)) + \mathbf{b}_{\text{t}} \mathbf{b}_{\text{t}}(1-d)$; $\det(J) = \exp(2\mathbf{a}_{\text{t}})$ (non-volume-preserving). / 仿射耦合, 非体积保持, 表达能力更强。

Q14: MAF vs IAF / 掩码自回归流与逆自回归流

Problem (问题): When would you choose MAF over IAF?
Solution (解答): Choose MAF when you need efficient density evaluation (training by MLE). Since its inverse ($\mathbf{x} \rightarrow \mathbf{z}$) is parallel. Choose IAF when you need efficient sampling (generation), since its forward ($\mathbf{z} \rightarrow \mathbf{x}$) is parallel. / MAF适合训练 (需要密度估计), IAF适合生成 (需快速采样)。

Q15: Camera Projection Matrix / 相机投影矩阵

Problem (问题): Write the full projection from world coordinates to pixel coordinates.
Solution (解答): $\mathbf{p}(\text{pixel}) = K [R | t] X$ (world) where $K = \begin{bmatrix} f & 0 & c_x \\ 0 & f & c_y \\ 0 & 0 & 1 \end{bmatrix}$, $[R | t]$ = 3×3 rotation, t = 3×1 translation (extrinsics). / 内参矩阵 K , 旋转 R 和平移 t 组成完整投影。

End of CS5187 Review Material / CS5187复习资料完

Q16: WGAN vs. Original GAN / WGAN与原始GAN对比

Problem (问题): Why is Wasserstein distance preferable to Jensen-Shannon divergence as a GAN training objective?

Solution (解答): - Jensen-Shannon divergence: When p_{data} and p_{G} have non-overlapping supports (common in high dimensions), D_{JS} = $\log 2$ is constant gradient provides no useful learning signal. / 当分布不重叠时JS散度为常数, 梯度为零, 无法提供学习信号。 - Wasserstein distance: Provides a meaningful, continuous gradient even when distributions don't overlap. The loss correlates with sample quality, enabling more stable training. / 即使分布不重叠也提供有意义的连续梯度, 损失与样本质量相关。 - Practical benefit: WGAN loss decreases as sample quality improves; can be used as a training monitor. / WGAN损失随样本质量提升而下降, 可作为训练监控指标。

Q17: CycleGAN Cycle Consistency / CycleGAN循环一致性

Problem (问题): Why is cycle consistency necessary in CycleGAN? What happens without it?

Solution (解答): - Without cycle consistency: The adversarial losses alone ensure that $G(X)$ looks like domain Y and $F(Y)$ looks like domain X , but there is no guarantee that $F(G(X)) \approx X$. The generators could map all inputs to a single mode in the target domain (mode collapse). / 仅靠反对数损失无法保证往返一致。生成器可能将所有输入映射到目标域的单一模式。 - With cycle consistency: $F(G(X)) \approx X$ and $G(F(Y)) \approx Y$ enforce that the translation preserves information content – the round trip must recover the original. / 循环一致性迫使翻译保留信息内容, 往返必须恢复原始输入。 - Loss: $\mathcal{L}_{\text{cyc}} = \mathbb{E}[\| F(G(X)) - X \|_1] + \mathbb{E}[\| G(F(Y)) - Y \|_1]$. / 循环一致性损失用L1距离衡量往返误差。

Q18: EBM Gradient Interpretation / EBM梯度解释

Problem (问题): Explain the positive phase and negative phase in EBM maximum likelihood training.

Solution (解答): $\nabla_{\theta} \log p_{\theta}(\mathbf{x}_{\text{data}}) = \nabla_{\theta} f_{\theta}(\mathbf{x}_{\text{data}})$ (positive phase) - $\mathbb{E}_{\mathbf{x} \sim p_{\theta}} [\nabla_{\theta} f_{\theta}(\mathbf{x})]$ (negative phase) - Positive phase (正相位): Pushes up f_{θ} at the training data location, making training data more probable. / 在训练数据位置推高 f_{θ} , 提高训练数据概率。 - Negative phase (负相位): Pushes down f_{θ} at locations where the model currently assigns high probability, reducing probability of non-data regions. / 在模型当前赋予高概率位置拉低 f_{θ} , 降低非数据区域的概率。 - Contrastive divergence: Approximates the negative phase with MCMC samples: $\nabla_{\theta} f_{\theta}(\mathbf{x}_{\text{data}}) - \nabla_{\theta} f_{\theta}(\mathbf{x}(\text{sample}))$. / 用MCMC采样近似负相位。

Q19: Score Function in EBMs / EBM中的分数函数